

Zijun James Chen

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EDUCATION

University of Technology Sydney

Ultimo, NSW

Bachelor of Computer Science (Honours) – AI & Data Analytics; Sub-major: AWS Industry 2023 – Mid 2027

- **WAM:** 84.74 (Distinction) | **GPA:** 6.48 / 7.00 | **Coursework:** NLP, Advanced Data Visualisation, Games Programming, AI Capstone

RESEARCH

Honours Research – Diffusion-Based Deepfake Video Detection

2026 – Present

University of Technology Sydney, School of Computer Science

Methodology in progress

- Investigating whether **audio-visual disharmony cues** (lip-speech misalignment) remain reliable for detecting deepfakes produced by modern **diffusion models**, compared with older GAN-based fakes
- Designed a **cross-dataset evaluation**: training a lip-audio cross-attention detector (DF-TransFusion-style) on **FakeAVCeleb** (GAN era) and testing on **DigiFakeAV** (diffusion era) to measure robustness degradation
- Built Python/**ffmpeg** data pipelines and identified pitfalls that inflate reported accuracy, such as **leading-silence artifacts** in fake audio acting as a detection shortcut, and class imbalance

PROJECTS

Multimodal Emotion Recognition | *PyTorch, MAE-DFER, timm, Transformers (team AI capstone)*

2026

- Team system predicting 7 emotion dimensions from video, audio and text on **CMU-MOSEI**; the fused model reached an average **CCC of 0.487**
- Owned the **video modality**: selected **MAE-DFER** over VideoMAE for its facial pre-training, and added attention pooling over backbone features
- Got an incomplete research codebase running by patching an unimplemented model branch, pinning **timm** to resolve version conflicts, and fixing dtype mismatches in the fusion layer

OOP Tutoring Chatbot | *Python, Phi-4-mini, QLoRA, FAISS, Flask (team project)*

2026

- Team-built educational chatbot that answers object-oriented programming questions, grounded in course material via **retrieval-augmented generation (RAG)** with a **FAISS** vector index
- Fine-tuned **Phi-4-mini-instruct** with **QLoRA** for efficient training on limited GPU memory, served through a **Flask** backend

VisNet – Lip-Reading Web Platform | *Python, PyTorch, JavaScript, HTML/CSS (team project)*

May 2025

- Team-built visual speech recognition system that transcribes speech from silent video, using a 3D ResNet-18 front-end with a Conformer encoder and Transformer decoder
- Fine-tuned from LRS3 weights to a best **WER of 46.41%** on LRS2, served through a web upload front-end with local inference

Stealth Game – Enemy AI & Gameplay Systems | *C++, Unreal Engine 5, Blueprints*

2026

- Developing enemy AI for a 3D stealth game in C++, including custom **A* pathfinding** for enemy navigation and in-engine debug tooling to visualise paths
- Built a weapon pickup and equip system with a C++ class hierarchy exposed to Blueprints, driving animation blending on the player character

EXPERIENCE

Plumbing & Tiling Handyman

Jan 2024 – June 2026

Union Pacific NSW Pty Ltd

Chipping Norton, NSW

- Completed 30+ residential plumbing and tiling jobs in a small team, planning installation sequences to stay on schedule and liaising directly with homeowners

Election Issuing Officer

Sep 2024

Liverpool City Council – NSW Local Government Elections

Hammondville, NSW

- Issued ballots and handled 1,500+ confidential documents; counted and validated 800+ ballot papers after close of polls

TECHNICAL SKILLS

Languages: Python, C++, C#, Java, SQL, JavaScript, HTML/CSS, Lua

ML & Data: PyTorch, Hugging Face Transformers, timm, QLoRA/PEFT, FAISS, RAG, Tableau

Tools & Platforms: Git, AWS SageMaker, Google Colab, Flask, ffmpeg, Unreal Engine 5, Unity, Visual Studio, VS Code